

KOBON TRIANGLES

How many triangles can n lines make?

Methods, results and what we are trying next · $n = 18$ and $n = 39$

93

$n = 18$ · ties the best known

470

$n = 39$ · official AutoLab score

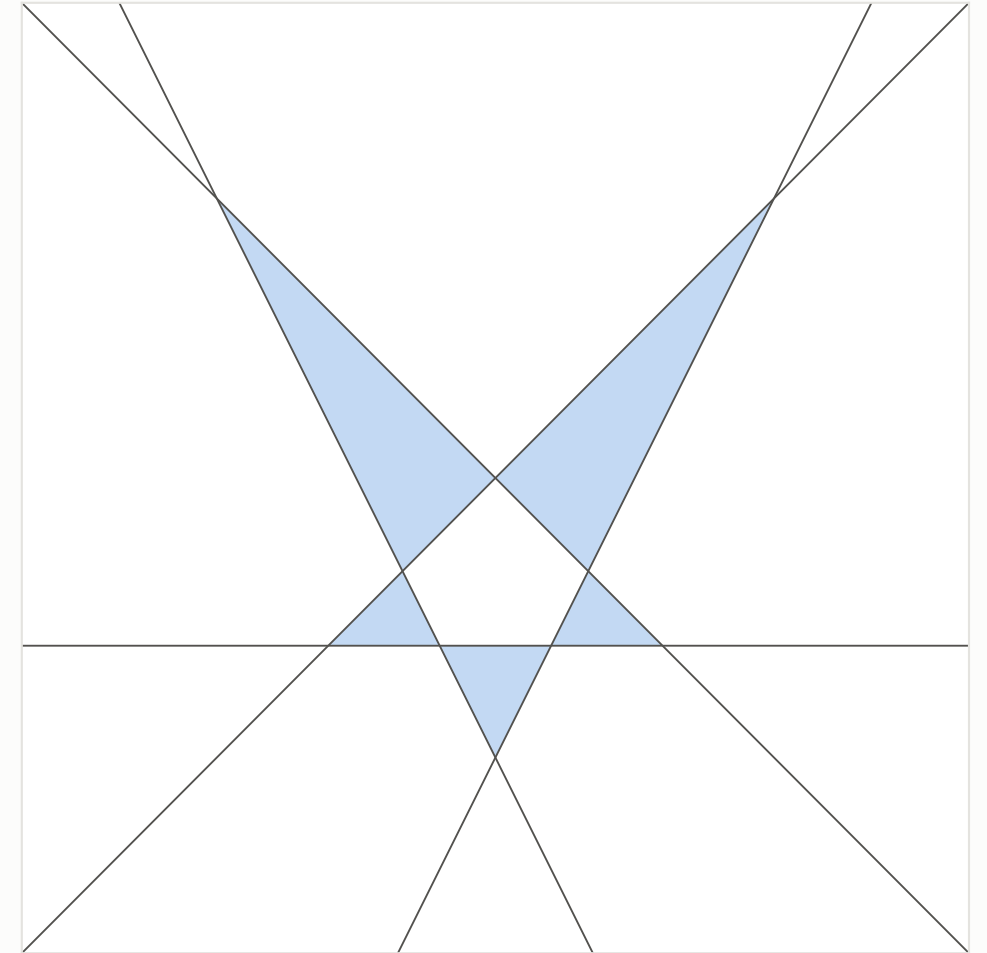
481

$n = 39$ · proven upper bound

The problem in one picture

- Draw exactly n straight lines.
- Count the triangles they cut out: closed 3-sided regions no other line passes through.
- A big triangle split by a line does not count.
- Goal: as many as possible — the Kobon triangle problem (still open for most n).
- Contest input: integers $a \cdot x + b \cdot y + c = 0$ per line, checked with exact arithmetic.

5 lines $\rightarrow$ 5 triangles (the maximum for 5)



The pieces that matter besides the triangle count

Crossing point

Where two lines meet. n lines in general position have $n(n-1)/2$ of them (153 for $n=18$, 741 for $n=39$).

Simple arrangement

No parallel lines and no point where three or more lines meet. Most theorems (and our SAT encoding) assume this.

Triple point

Three lines through one point. Merges crossings; lets one segment border two triangles. Our 470 has 5 of them.

Parallel pair

Two lines that never meet (they meet 'at infinity'). Allowed by the contest; one known 93 for $n=18$ uses 4 parallel pairs.

Bounded segment

A piece of a line between two consecutive crossings. Each triangle uses 3 of them; unused ones are 'wasted' capacity.

Upper bound (Tamura)

Each line has at most $n-2$ segments, so triangles $\leq \lfloor n(n-2)/3 \rfloor$: 96 for $n=18$, 481 for $n=39$ (hitting it = 'perfect').

Counting identity checked on every result:

$$3 \times \text{triangles} = (\text{segments used once}) + 2 \times (\text{segments shared by two triangles})$$

n = 18: we match the best known result, 93

93

ours (official score)

93

best known in the literature

96

proven upper bound (95 claimed, 2007 draft)

- Blanc (2011) proved: a SIMPLE arrangement of 18 lines has at most 93 triangles.
- So 94 would need triple points or parallel lines, and nobody has found one
 - (the Parpalak–Utkin gallery lists 3,016 different 93s and no 94).
- Value for the competition: matching a known result earns no new-math credit;
 - it confirmed our end-to-end pipeline (search → exact check → official evaluation).

Five different ways people reach 93 for $n = 18$

A · Annealing (ours)

Start from random lines, nudge one line at a time, keep changes that don't lose triangles, occasionally accept a worse one to escape dead ends. Found 93 in ~1 minute on 10 cores (7 of 10 runs).

B · Symmetric construction

Bader (2007): design the lines with 3-fold rotational symmetry, so you only choose 6 lines and rotate them. Fewer choices, cleaner structure.

C · SAT + straightening

Savchuk (2025): describe only the ORDER of crossings as true/false variables, let a SAT solver find a valid pattern, then bend it into straight lines.

D · Exhaustive search

Parpalak & Utkin (2026): enumerate combinatorial arrangements, then fit lines by linear programming. Their gallery has 3,016 different 93s.

E · Machine-checked proof

A 93 whose count was proved inside the Lean theorem prover (yhinai/openmath) — the triangle count itself is formally verified, not just computed.

Why they all stop at 93

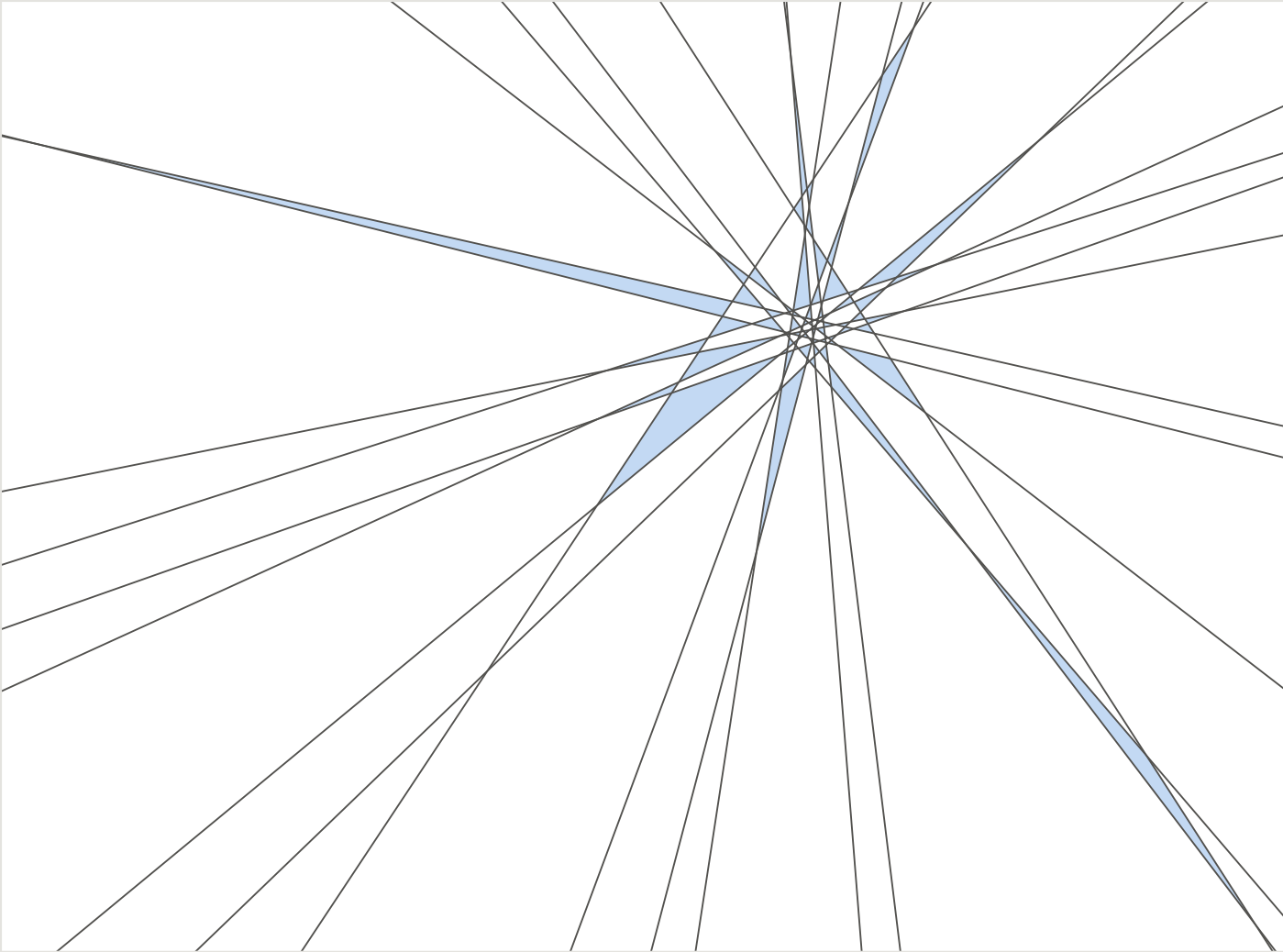
Blanc (2011): simple 18-line arrangements can't beat 93. Beating it needs triple points or parallels — tried in thousands of gallery variants, never above 93.

Same 93 triangles, very different structures

Solution	Triangles	Crossings	Triple pts	Parallel	Segments	Unused	Shared	Digits
Ours: simulated annealing	93	153	0	0	288	9	0	8
Gallery: 4 triple points	93	145	4	0	276	5	8	7
Gallery: 3 triple pts + 1 parallel pair	93	146	3	1	277	4	6	12
Gallery: 4 parallel pairs	93	149	0	4	280	1	0	18
Lean-verified (yhinaï/openmath)	93	152	0	1	286	7	0	6

- Ours is simple: 153 crossings, 288 segments, 9 unused — the maximum a simple 18-line arrangement allows.
- Triple points and parallels 'spend' crossings but waste fewer segments (down to 1 unused).
- Digits = size of the largest integer coefficient; all fit the contest's 10^{30} limit.

Our 93-triangle arrangement for $n = 18$



- 18 lines, 93 triangles (shaded)
- 153 crossing points, all simple
- 288 bounded segments,
 - 279 used, 9 unused
- Integer coefficients ≤ 8 digits
- Official score 93 (AutoLab, 27 Sep)
- Shaded = the 93 triangles; the view
 - is fitted to them (lines extend further).

n = 39: 470 triangles, officially scored

470

ours, official (27 Sep)

450

published 38-line base it extends

481

proven upper bound

?

published value (OEIS)

- No 39-line arrangement was published before this event — OEIS lists $a(39)$ as '?'.
 - The bound $481 = 39 \cdot 37/3$ holds for every arrangement (Felsner-Kriegel 1999);
 - reaching it needs a 'perfect' arrangement: every segment used by exactly one triangle.
- The only thing ruled out so far: a perfect arrangement with 13-fold symmetry (Savchuk 2025).
- Gap to close: 11 triangles between what we can draw (470) and what is provably impossible (482+).

How we got 470

1 Start

Parpalak-Utkin gallery:
248 different 38-line
arrangements, 450
triangles each (best known)

2 Add one line

For each, try every
position of a 39th line
(angle × gap grid, scored
incrementally in ~6 s)

3 Exact check

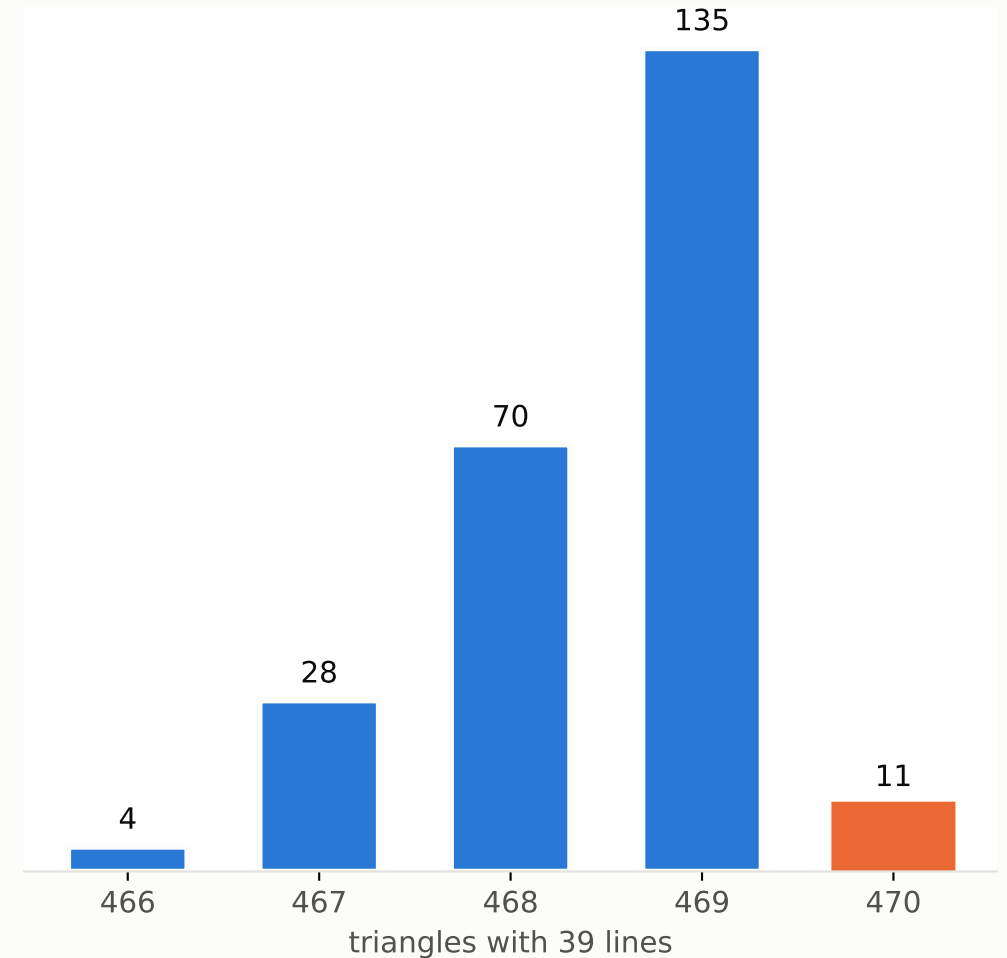
Re-count the winner with
integer arithmetic, then
the contest's own evaluator
(two independent counters)

4 Submit

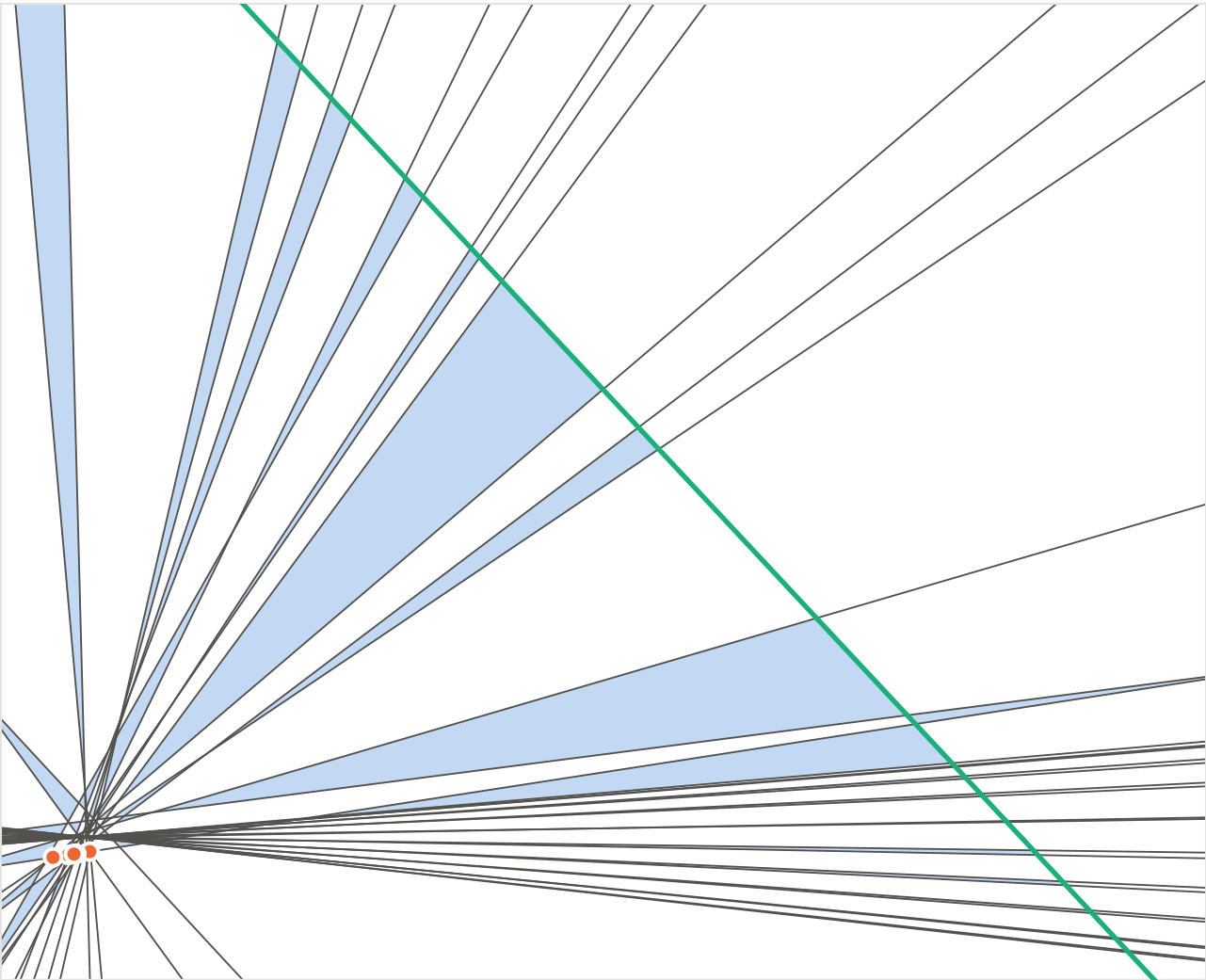
Official AutoLab climb:
locked experiment, run
on our Mac node, scored
470 at n = 39

11 of 248 bases reach 470; none reach 471 —
checked over every combinatorial position of the new line.

Best triangles after adding one line (248 bases)



Anatomy of our 470



	38 base	Our 39
Lines	38	39
Triangles	450	470
Crossing points	693	731
Triple points	5	5
Bounded segments	1353	1428
Unused segments	13	28
Shared by 2 triangles	10	10
Upper bound	456	481

- Green = the line we added (+20 triangles).
- Orange = triple points (inherited from the base).
- Picture: zoom on the new line's triangles — the full arrangement spans many scales.
- Made simple (no triple points) it drops to 466–468.

What we tried that did not beat 470 (and what it taught us)

Local search

Remove 1 or 2 lines and put back the best ones: ~4,700 moves on three 470s, never above 470. The 470s are strict local optima.

Line through a crossing

Force the new line exactly through an existing crossing (makes a triple point). Best 469 — worse than a free line.

Symmetric SAT (3-fold)

Solver found curved-line patterns with 475, 477, 480 and 480 triangles — but none could be straightened (best straight fit broke 339–477 crossing orders).

Lesson: 'stretchability'

Curved-line (pseudoline) patterns are easy to find; only a tiny, shrinking fraction can be drawn with straight lines (~99% at 19 lines, ~1% at 27; 0 of 46 perfect 39-line patterns we tried).

Last approach tried: SAT neighbourhood search

Freeze

Take a straight 470-type arrangement (made simple: 466–468). Freeze all lines except 3–6 'free' ones near wasted segments.

Ask SAT

Encode every crossing order as one bit per triple of lines (9,139 bits). Ask: can the free lines be re-threaded so at most $1443 - 3T$ segments are wasted, for $T = \text{best} + 1$?

Straighten

Only the free lines move, so straightening is small: pick their slopes, then solve a linear program for their positions.

Verify

Exact integer re-count; if it beats the best, it becomes the new base and the target rises ($469 \rightarrow 470 \rightarrow 471 \dots$).

- Why this can work where symmetric SAT failed: the answer stays close to a picture we can already
 - draw, so straightening is a small, well-posed problem instead of a shot in the dark.
- Outcome: 394 neighbourhoods (8 cores, ~3 h): 8 curved-line 469s, none straightenable; 7 local 'no' answers;
 - the rest timed out. A parallel 2–4-line re-insertion beam (~2,000 moves) also stopped at 470.

Where the maximum for $n = 39$ stands

470
ours (drawn)

481
proven maximum

open: 471 – 481

- Proven: nothing can beat 481 (true for all arrangements, including triple points and parallels).
- Not yet known for us: whether 481 (or anything above 470) can be drawn with straight lines.
- Tried without success: SAT neighbourhood search (394) and 2-4-line re-insertion (~2,000 moves).
 - Open directions: new global constructions; straightening many more near-perfect curved-line patterns.

How every number here was checked

- Exact arithmetic everywhere: lines are integers, crossings are exact fractions — no floating point in any count.
- Two independent counters (ours + the contest evaluator's code) plus the counting identity on every result.
- Official scores come only from AutoLab's own evaluator (signed report, hill tree hash 7d3f1d91).
- Credit: the 38-line bases and several 93s are Parpalak & Utkin's published gallery; the symmetric 93 is Bader's; the SAT tooling is Savchuk's (kobon-cnf, LineOrder) with the Kissat solver.
- Ours: the search and insertion pipeline, the 39-line results, the SAT neighbourhood method.
- Tools: Python (numba, python-sat, SciPy/HiGHS), Kissat 4.0.4, Claude Code (Claude Opus 5.5) as the assistant.